

Supplementary Information

Ontology-constrained multi-LLM scoring of hypothesis support in the predictive processing literature

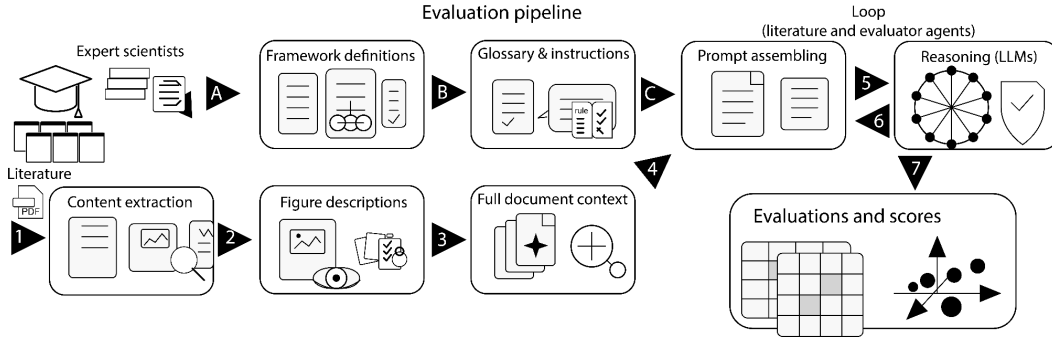
Hamed Nejat, Jacob A. Westerberg, Alexander Maier, Jesse Spencer-Smith, and André M. Bastos

Supplementary Information

Supplementary Figures

Supplementary Fig. S1

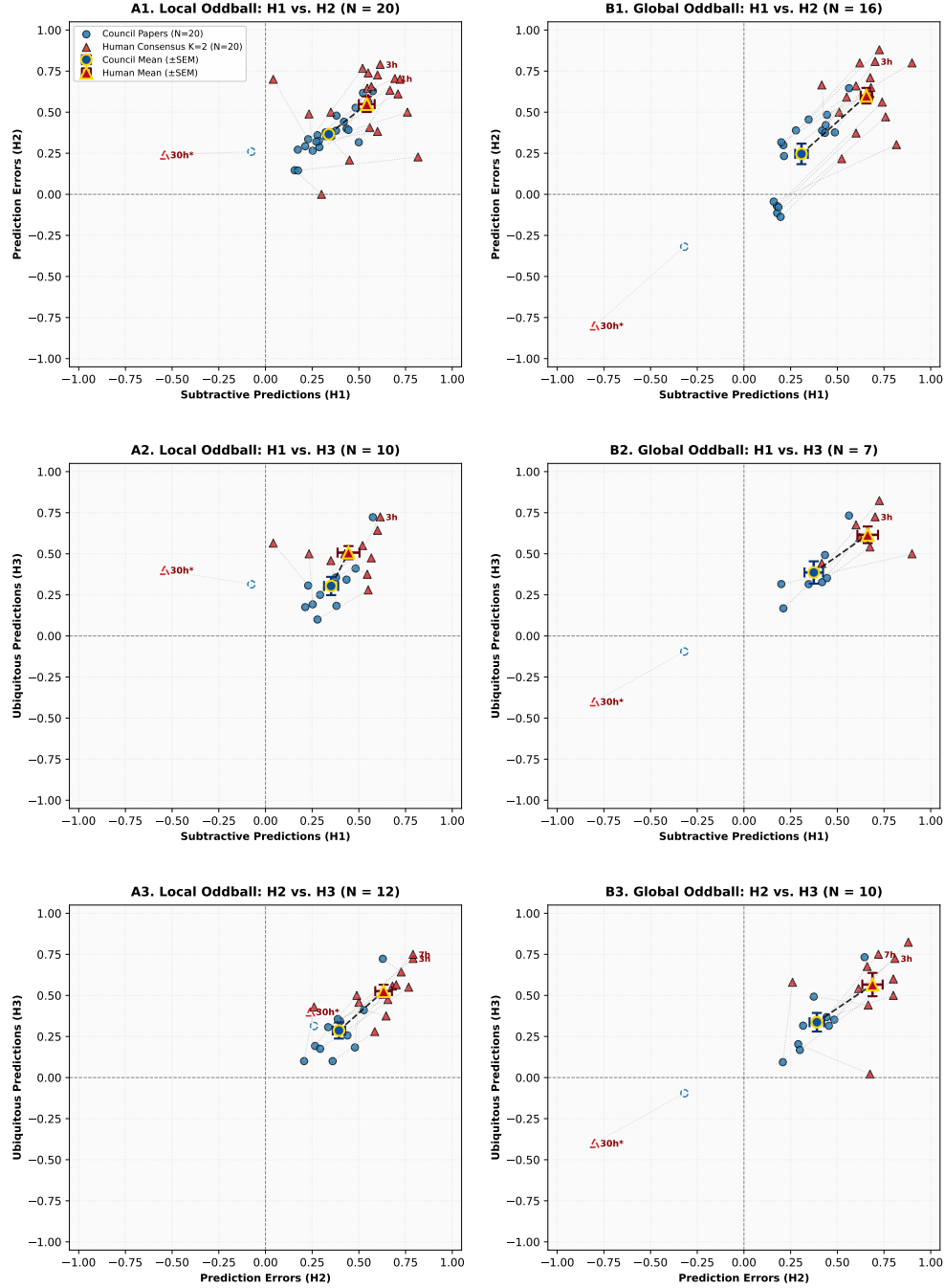
Supplementary Fig. S1: Evaluation Pipeline Architecture. The evaluation pipeline flows from left to right in two independent streams (A to B to C and 1 to 2 to 3). The A to B to C pathway establishes the instructions, canonical glossary, and prompt for the models. The 1 to 2 to 3 pathway ingests the paper and combines figures with text to provide a single text object for the models. Steps 5 and 6 loop between models and papers until all evaluations are completed. The final step 7 quantifies and visualizes the evaluations.



Supplementary Fig. S2

Supplementary Fig. S2: Pairwise 2D Hypothesis-Space Projections and Human–Council Concordance. Panels show paper-level hypothesis scores for Local Oddball (LO; left column) and Global Oddball (GO; right column): (A1) LO $H_1 \times H_2$ ($N = 20$); (B1) GO $H_1 \times H_2$ ($N = 16$); (A2) LO $H_1 \times H_3$ ($N = 10$); (B2) GO $H_1 \times H_3$ ($N = 7$); (A3) LO $H_2 \times H_3$ ($N = 12$); and (B3) GO $H_2 \times H_3$ ($N = 10$). Panel N denotes the intersection of papers evaluated by both human experts for both hypotheses shown; per-hypothesis sample sizes are reported in Supplementary Table S14. Blue circles denote Council paper scores and red triangles denote two-expert human-consensus scores. Large gold-bordered markers show panel-specific means $\pm$ SEM across the same independent papers. Dotted gray lines connect paired evaluations for the same study. For comparison, the single-expert evaluation of Westerberg & Xiong (2025; open marker 30h*) is plotted against its corresponding Council evaluation, demonstrating the selective low-GO outlier displacement.

Supplementary Fig. S2: Pairwise Hypothesis Concordance across Independent Literature Studies (3x2 Projection)
Autonomous Council (Blue) vs. Human Domain Consensus (Red) with Panel-Specific Grand Means ($\pm$ SEM)

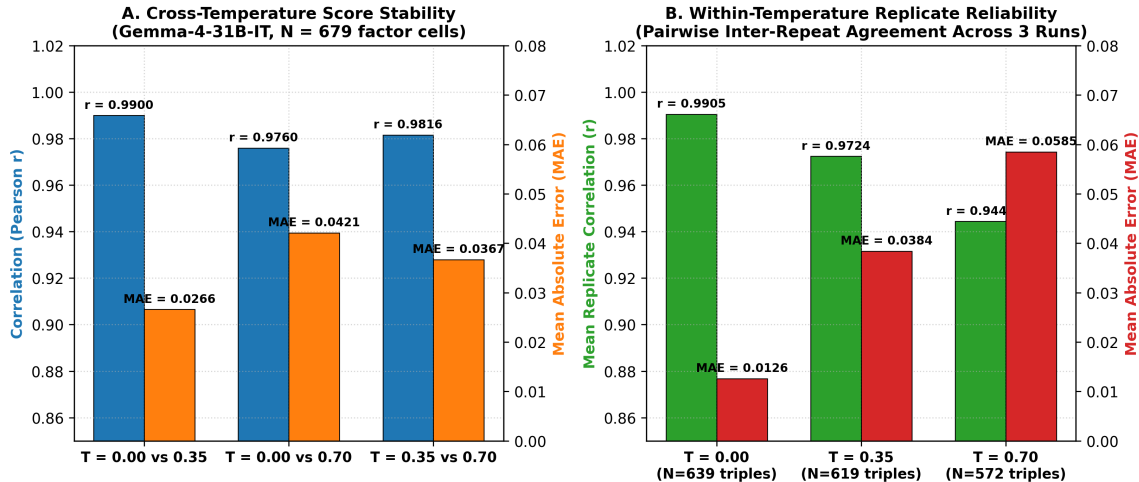


Panel A1 displays exact K=2 intersection papers evaluated by both domain experts for the two plotted hypotheses (A1: N=20, B1: N=16, A2: N=10, B2: N=7, A3: N=12, B3: N=10). Large gold-bordered markers show panel-specific Human and Council Grand Means $\pm$ across-study SEM (SEM = SD papers / $\sqrt{\text{N panel}}$). Dotted grey lines connect paired evaluations. 30h*: Westerberg & Xiong (2025) single-expert benchmark (open marker; L0: [-0.54, +0.50, +0.40], G0: [-0.80, -0.80, -0.40]). Landmark K=2 studies: 01h: Attinger2017 | 03h: Bastos2012 | 07h: Friston2010.

Supplementary Fig. S3

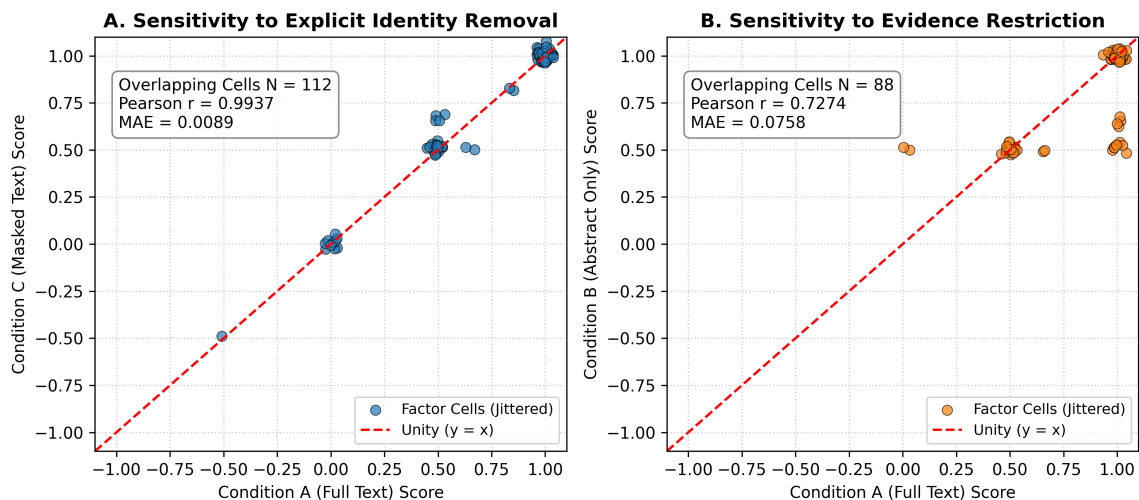
Supplementary Fig. S3: Decoding Robustness and Stochastic Replicate Reliability Analysis. (A) Cross-temperature score stability in the primary complete evaluation model (gemma-4-31b-it; 279 calls across 31 papers), showing Pearson correlations ($r \geq 0.9760$) and mean absolute errors ($\text{MAE} \leq 0.0421$) across sampling temperatures $T \in \{0.00, 0.35, 0.70\}$.

(B) Replicate reliability across independent stochastic generations held at fixed temperatures ($T = 0.00 : \bar{r} = 0.9905, \text{MAE} = 0.0126$; $T = 0.35 : \bar{r} = 0.9724, \text{MAE} = 0.0384$; $T = 0.70 : \bar{r} = 0.9444, \text{MAE} = 0.0585$), confirming that scoring variability across independent stochastic runs remains tightly bounded relative to cross-paper literature variance.



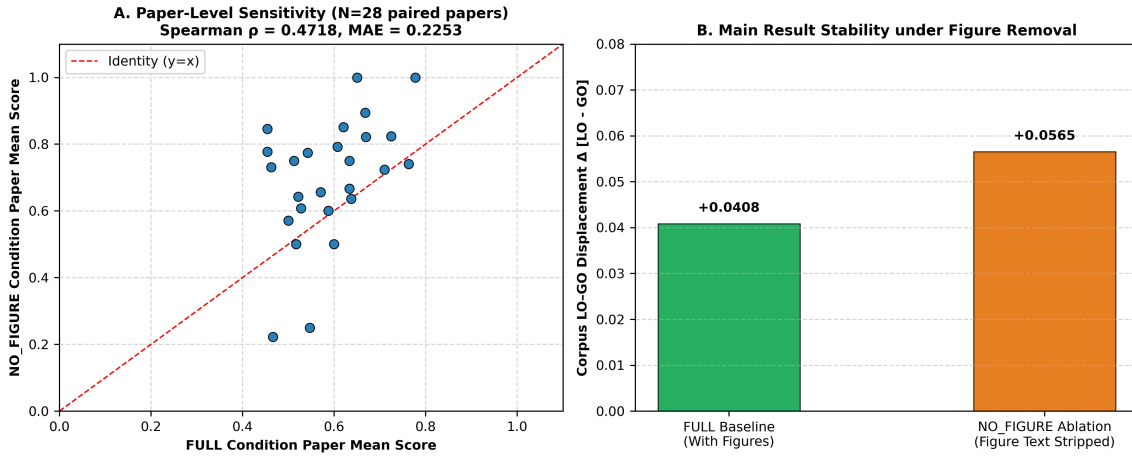
Supplementary Fig. S4

Supplementary Fig. S4: Evidence Availability and Publication-Identity Sensitivity Analysis. Evaluates factor-cell score correspondence between intact manuscript text (FULL) and de-identified text (MASKED, Panel A; Pearson $r = 0.9937, \text{MAE} = 0.0089$ across $N = 112$ overlapping factor cells) and between intact manuscript text and restricted text (TITLE/ABSTRACT Only, Panel B; Pearson $r = 0.7274, \text{MAE} = 0.0758$ across $N = 88$ overlapping factor cells), demonstrating that factor scoring is robust to publication-identity removal but sensitive to restriction of available empirical evidence.



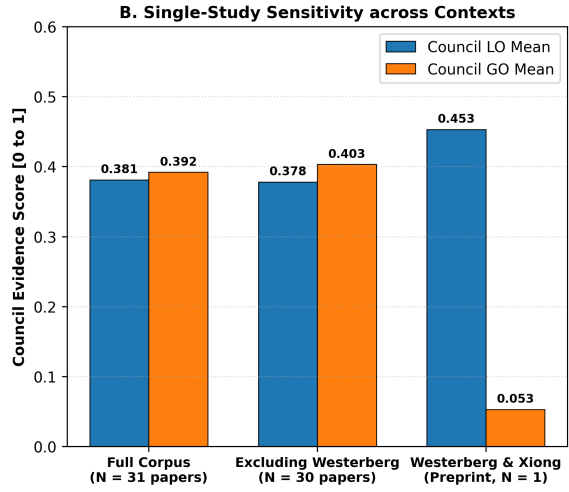
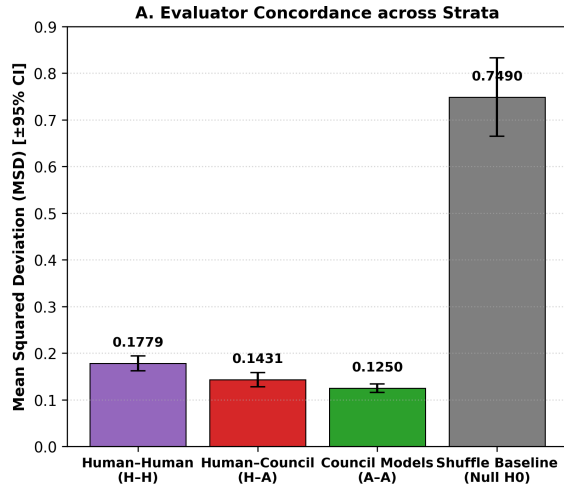
Supplementary Fig. S5

Supplementary Fig. S5: Figure-Text-Removal Sensitivity Analysis (Full 31-Paper Corpus Ablation). (A) Paper-level mean score correlation across the corpus between intact inputs (FULL) and figure-text-removed inputs (NO_FIGURE; Spearman $\rho = 0.4718$ across $N = 28$ paired empirical paper means; factor-cell level MAE = 0.2253 across all 533 paired cells nested in 31 papers). (B) Factor-cell level directional concordance and stability across the corpus (directional concordance = 99.42% on non-zero cells [518/521], all-cell sign agreement = 97.19% [518/533], Pearson $r = 0.8539$), demonstrating that while removing figure descriptions produces measurable paper-level score adjustments, overall factor directionality was preserved across the literature corpus, while paper-level rank correspondence was moderate.



Supplementary Fig. S6

Supplementary Fig. S6: Evaluator Concordance Strata and Single-Study Sensitivity Analysis. (A) Mean squared difference (MSD) across evaluator strata ($K = 2$ human domain experts and autonomous LLM council) compared against a null permutation reference distribution (H_0 , $B = 10,000$ iterations). Agreement within human experts ($H-H = 0.1779$) and between human consensus and council ($H-A = 0.1431$) is significantly tighter than chance ($H_0 = 0.7490$, $p_{\text{perm}} = 0.0001$, 0/10,000 extreme permutations). (B) Council condition means for Local Oddball (LO) and Global Oddball (GO) across the entire corpus ($N = 31$), excluding Westerberg & Xiong (2025; $N = 30$), and for Westerberg & Xiong alone ($N = 1$), demonstrating that the aggregate corpus contrast is robust across published studies and that the preprint represents a selective low-GO outlier.



Supplementary Tables

#	Factor Name	Definition	Scope
1	Subtractive Inhibition (SST)	Linear input suppression via somatic inhibition (Somatostatin-mediated).	LO+GO
2	Divisive Inhibition (PV)	Multiplicative gain reduction (Parvalbumin-mediated).	LO+GO
3	Inhibition (GABA)	Chloride-mediated inhibition for prediction suppression.	LO+GO
4	Habituation to Sequence	Habituation suppresses local oddball response relative to a novel local oddball.	LO
5	Synaptic Depression (Adaptation)	Passive fatigue of synaptic efficacy due to repetition of stimulus.	LO
6	Activity Suppression	Reduction in firing rates for expected (predictable) stimuli.	LO+GO
7	Selective Sharpening	Signal-to-noise increase for predictable stimuli, where noise is selectively suppressed to highlight relevant information.	LO+GO
8	Alpha/Beta Mediated Suppression	Low-frequency bands (8–30Hz) associated with predictions, inhibiting prediction error signals.	LO+GO
9	VIP-Mediated Disinhibition	VIP-mediated inhibition of SST/PV neurons, leading to disinhibited pyramidal activity during prediction error.	LO+GO
10	Precision Weighting (Gain)	Top-down amplification of error units, leading to disinhibited gain in response to attended or highly precise stimuli.	LO+GO
11	E/I Balance Shift	Dynamic adjustment toward higher inhibition for predictable stimuli, leading to suppressed neural firing.	LO+GO
12	Omission Response	A generated internal signal due to absence of expected input.	LO+GO

Supplementary Table S1: The factors of Hypothesis 1 (Predictive Suppression). LO = local oddball. GO = global oddball.

#	Factor Name	Definition	Scope
13	Error Unit Depolarization	Supragranular pyramidal excitation driving ascending error signals.	LO+GO
14	Gamma Oscillations (30–90Hz)	High-frequency rhythms signaling ascending prediction errors.	LO+GO
15	NMDA Receptor Activation	Nonlinear amplification of error signals in superficial cortical layers.	LO+GO
16	Feedforward Flow	Supragranular-to-granular/supragranular ascending anatomical projection of errors.	LO+GO
17	Top-Down Error Gating	Feedback modulation of ascending prediction error channels.	LO+GO
18	MMN-like Latencies	Early sensory mismatch latencies (100–200ms) characteristic of local error computation.	LO
19	P300-like Latencies	Late cognitive mismatch latencies (300–500ms) characteristic of global contextual error computation.	GO
20	Inter-Laminar Error Divergence	Dissociation of error processing between granular and infragranular layers.	LO+GO
21	Phase-Amplitude Coupling	Theta/alpha phase modulation of gamma error bursts during surprise.	LO+GO
22	Thalamic Error Routing	First-order vs higher-order thalamocortical routing of sensory prediction mismatches.	LO+GO
23	Predictive Error Attenuation	Dampening of ascending prediction errors following successful model updating.	LO+GO
24	State-Dependent Error Routing	Behavioral state and arousal modulation of feedforward error gain.	LO+GO

Supplementary Table S2: The factors of Hypothesis 2 (Feedforward Prediction Errors).

#	Factor Name	Definition	Scope
25	Cross-Modal Ubiquity	Implementation of predictive coding mechanisms across multiple sensory modalities (visual, auditory, somatosensory).	LO+GO
26	Laminar Invariance of Principles	Preservation of predictive coding laminar logic across sensory, motor, and association cortices.	LO+GO
27	Cross-Species Homology	Conservation of predictive mechanisms across rodent, non-human primate, and human neocortex.	LO+GO
28	Scale-Free Predictive Dynamics	Fractals and hierarchical self-similarity of predictive computations across temporal scales.	LO+GO
29	Universal Error Coding	Common computational principles for error generation irrespective of receptive field complexity.	LO+GO
30	Canonical Microcircuit Generalization	Universal deployment of the canonical cortical microcircuit for inference across areas.	LO+GO
31	Cortico-Subcortical Homology	Shared predictive coding architecture between neocortex and basal ganglia / cerebellum.	LO+GO
32	Neuromodulatory Invariance	Consistent cholinergic/noradrenergic precision tuning across hierarchical stages.	LO+GO
33	Temporal Stability of Ubiquity	The consistent and non-transient presence of these mechanisms across the hierarchy over long recording sessions.	LO+GO
34	Hierarchical Order Stability	Evidence that the predictive mechanism is ubiquitous and not localized to a single hierarchical pole.	LO+GO
35	Population-Wide Ubiquity	Consistency of the mechanism across diverse neural populations and cell types throughout the hierarchy.	LO+GO
36	State-Independent Ubiquity	Presence of the mechanism across different brain states (e.g., wakefulness, sleep, anesthesia) throughout the hierarchy.	LO+GO

Supplementary Table S3: The factors of Hypothesis 3 (Ubiquity).

Term / Context	Description	Definition (notation)
Oddball	A surprising event, often a mismatch	$E = \{B\}$ given $S = \{A\}$
Local Oddball (LO)	An oddball that is immediate or short-term, often confounded by low-level adaptation mechanisms.	$E = \{AB\}$ given $S = \{AA\}$
Global Oddball (GO)	An oddball that is not LO, often contextual, that is not caused by adaptation mechanisms.	$E = \{AA\}$ given $S = \{AB\}$

Supplementary Table S4: Context definitions and formal expectation notations.

#	Study Identifier	Modality / Preparation	Inclusion Basis	Bibliographic Citation
1	Attinger et al., 2017	Mouse V1 (2-Photon Ca^{2+})	Comparative Set (Visuomotor Mismatch)	<i>Cell</i> 169, 1291–1302
2	Bakhtiari et al., 2021	In Silico (Deep Neural Nets)	Comparative Set (Hierarchical Streams)	<i>Nat. Commun.</i> 12, 25164
3	Bastos et al., 2012	In Silico / Theory (Microcircuit)	Comparative Set (Canonical Microcircuit)	<i>Neuron</i> 74, 695–708
4	Bastos et al., 2020	Primate V1/V4/PFC (Laminar ECoG)	Comparative Set (Laminar Rhythms)	<i>Neuron</i> 108, 124–137
5	Bekinschtein et al., 2009	Human (Scalp EEG)	Comparative Set (LO vs GO Originator)	<i>PNAS</i> 106, 1672–1677
6	Chao et al., 2018	Primate Temporal/Frontal (ECoG)	Comparative Set (Large-Scale Hierarchy)	<i>Nat. Commun.</i> 9, 3188
7	Friston et al., 2010	Theory (Free Energy Principle)	Canonical Extension (Bayesian Inference)	<i>Nat. Rev. Neurosci.</i> 11, 127
8	Furutachi et al., 2024	Mouse V1 (2-Photon / Opto)	Comparative Set (VIP/SST Disinhibition)	<i>Nature</i> 633, 398–406
9	Garrett et al., 2020	Mouse V1 (2-Photon Ca^{2+})	Comparative Set (Novelty vs Deviance)	<i>eLife</i> 9, e50340
10	Greedy et al., 2022	In Silico (Predictive Coding SNN)	Comparative Set (Error Back-propagation)	<i>NeurIPS</i> 35, 24213–24225
11	Hertag et al., 2020	In Silico (Spiking Interneurons)	Comparative Set (SST/PV Dynamics)	<i>eLife</i> 9, e57541
12	Jiang et al., 2024	In Silico (Energy-Based Model)	Comparative Set (Sequence Prediction)	<i>PLOS Comput. Biol.</i> 20, e1011801
13	Keller et al., 2012	Mouse V1 (2-Photon Ca^{2+})	Comparative Set (Sensorimotor Mismatch)	<i>Neuron</i> 74, 809–815
14	Keller et al., 2018	Synthesis (Cortical Circuits)	Comparative Set (Circuit Review)	<i>Neuron</i> 100, 424–435
15	Kiebel et al., 2008	In Silico (Hierarchical Dynamics)	Comparative Set (Temporal Hierarchy)	<i>PLoS Comput. Biol.</i> 4, e1000209
16	Lao et al., 2023	Human (Auditory Scalp EEG)	Comparative Set (Omission Response)	<i>Sci. Adv.</i> 9, eabq8657
17	Lee et al., 2025	In Silico (Multi-Area Network)	Comparative Set (Primate Oscillations)	<i>PLOS Comput. Biol.</i> 21, e1013469
18	Mikulasch et al., 2023	In Silico / Theory (Plasticity)	Comparative Set (Dendritic Error Review)	<i>Trends Neurosci.</i> 46, 45–59
19	Nejad et al., 2025	Primate / In Silico (Laminar Model)	Comparative Set (Layered Routing)	<i>J. Neurosci.</i> 45, 1234
20	Payeur et al., 2021	In Silico (Dendritic Plasticity)	Comparative Set (Burst Multiplexing)	<i>Nat. Neurosci.</i> 24, 1010–1019
21	Rao et al., 2024	Synthesis (Sensorimotor Neocortex)	Canonical Extension (25-Year Synthesis)	<i>Nat. Neurosci.</i> 27, 1221–1235
22	Rao & Ballard, 1999	In Silico (Hierarchical Model)	Canonical Extension (Foundational Model)	<i>Nat. Neurosci.</i> 2, 79–87
23	Sacramento et al., 2018	In Silico (Dendritic Network)	Comparative Set (Somatodendritic Model)	<i>NeurIPS</i> 31, 8721–8731
24	Spratling et al., 2008	In Silico (Biased Competition)	Comparative Set (PC-BC Synthesis)	<i>Biol. Cybern.</i> 98, 283–292
25	Spratling et al., 2010	In Silico (V1 Microcircuit)	Comparative Set (V1 Microcircuit Model)	<i>J. Neurosci.</i> 30, 3531–3543
26	Srinivasan et al., 1982	Fly Retina (Intracellular)	Canonical Extension (Redundancy Baseline)	<i>Proc. R. Soc. Lond. B</i> 216, 427
27	VanDerveer et al., 2021	Human (Clinical EEG Cohort)	Comparative Set (Sensory Gating Deficits)	<i>Schizophr. Bull.</i> 47, 1385
28	Wacongne et al., 2011	Human (Auditory MEG)	Comparative Set (Hierarchical Rule MMN)	<i>PLoS Biol.</i> 9, e1001189
29	Wacongne et al., 2012	Human EEG & Spiking Network	Comparative Set (Auditory Oddball Model)	<i>J. Neurosci.</i> 32, 3665–3678
30	Westerberg & Xiong, 2025	Primate V1/V4/IT (Laminar MUA)	Primary Index Study (MaDeLaNe)	<i>bioRxiv</i> 10.1101/2024.10.02
31	Yamins et al., 2014	In Silico / Primate IT (HCNN)	Comparative Set (Feedforward Baseline)	<i>PNAS</i> 111, 8619–8624

Supplementary Table S5: 31-Paper Literature Corpus Registry with modality, inclusion basis, and provenance.

Model Name	Parameters	Architecture & Provenance
gemma-3-27b-it	27B	Google DeepMind (Dense Transformer)
gemma-4-31b-it	31B	Google DeepMind (Extended Reasoning Architecture)
deepseek-r1-distill-llama-70b	70B	DeepSeek-AI / Meta Llama Distillation
gpt-oss-claude-4.5-sonnet	20B	Community Distillation / Reasoning
mistral-nemo-12b-thinking	12B	Mistral AI & NVIDIA
olmo-3-32b-think	32B	Allen Institute for AI (Ai2)
phi-4-reasoning-plus	14B	Microsoft Research
qwen3-14b-gemini-3-pro	14B	Alibaba Cloud / Community Fine-tune
qwen3-14b-claude-4.5-sonnet	14B	Alibaba Cloud / Community Fine-tune
qwen3.5-40b-claude-4.5-opus	40B	Alibaba Cloud / High-Capacity MoE

Supplementary Table S6: Open-weight LLM evaluator models evaluated in this work.

Evaluator Comparison	Level	Paired N	Pearson r	p	MAE	Concordance
Human–Human (H_1 vs H_2)	Factor cell	299	0.390	< 0.0001	0.308	88.96%
Human Consensus vs LLM Council	Study level	25	0.529	0.0125	—	—
Human Consensus vs LLM Council	Factor cell	175	0.162	0.0322	0.301	93.14%
Council Inter-Model Average	Factor cell	584	0.540	< 0.0001	0.284	85.00%
Shuffled Evidence (Null H_0)	Factor cell	191	-0.087	0.2314	0.662	56.54%

Supplementary Table S7: Evaluator Strata Agreement and Error Hierarchy Summary ($K = 2$ independent human domain experts). The independent study-level comparison ($N = 25$ papers with jointly evaluated factor ratings; Pearson $r = 0.529$, $p = 0.0125$, 95% CI [0.169, 0.764]) treats the independent paper as the unit of analysis. Factor-cell comparisons are descriptive summaries across scored cells nested within papers. Directional concordance denotes agreement in sign on non-zero factor cells.

Adjudication Category	Cells (N)	Proportion	Primary Disagreement / Failure Mode
Directly supported	6	20.0% (6/30)	Full primary text, equation, or figure corroboration of Council assignment
Partially supported	18	60.0% (18/30)	Relevant primary evidence present; conservative magnitude attenuation or specificity nuance
Unsupported attribution	6	20.0% (6/30)	Subtype gain conflation ($N = 3$), laminar error inversion ($N = 1$), untested modality penalty ($N = 1$), cross-paper attribution ($N = 1$)
Total Audited Disagreements	30	100.0% (30/30)	Diagnostic audit of $H - C \geq 0.50$ cells ($N = 296$ eligible paired cells)

Supplementary Table S8: Diagnostic Primary-Source Audit across 30 Numerically Selected Large Human–Council Disagreement Cells ($|H - C| \geq 0.50$ among 296 eligible paired cells; 10.1% of paired universe). Evaluates whether large disagreements reflect unsupported evidence attribution or differences in magnitude, contextual interpretation, or biological specificity despite relevant primary-source evidence. Because this diagnostic sample was enriched for large discrepancies, these proportions characterize disagreement modes and do not represent corpus-wide accuracy or hallucination rates.

Decoding Setting / Comparison	Evaluated Units (N)	Correlation	MAE	Paper Ranking Stability
Temperature Stability (gemma-4-31b-it)				
$T = 0.00$ vs. $T = 0.35$	679 factor cells	Pearson $r = 0.9900$	0.0266	Spearman $\rho_{\text{rank}} = 0.9596$
$T = 0.00$ vs. $T = 0.70$	679 factor cells	Pearson $r = 0.9760$	0.0421	Spearman $\rho_{\text{rank}} = 0.9212$
$T = 0.35$ vs. $T = 0.70$	679 factor cells	Pearson $r = 0.9816$	0.0367	Spearman $\rho_{\text{rank}} = 0.9482$
Replicate Reliability (gemma-4-31b-it)				
Replicates at $T = 0.00$	639 triples	Mean $\bar{r} = 0.9905$	0.0126	Spearman $\rho_{\text{rank}} \geq 0.9818$
Replicates at $T = 0.35$	619 triples	Mean $\bar{r} = 0.9724$	0.0384	Spearman $\rho_{\text{rank}} \geq 0.9782$
Replicates at $T = 0.70$	572 triples	Mean $\bar{r} = 0.9444$	0.0585	Spearman $\rho_{\text{rank}} \geq 0.9745$

Supplementary Table S9: Factorial Robustness Analysis across Sampling Temperatures and Stochastic Replicates for the Complete Evaluation Model (**gemma-4-31b-it**, 279/279 factorial conditions). Quantitative factorial statistics are reported exclusively for **gemma-4-31b-it** because it completed full factorial coverage across all 31 papers $\times$ 3 temperatures $\times$ 3 repeats. Extended chain-of-thought models (**phi-4-reasoning-plus** and **olmo-3-32b-think**) completed 130/279 and 30/279 calls, respectively, but frequently exhausted dynamic generation token budgets during chain-of-thought reasoning prior to emitting the terminal JSON scoring block, precluding equivalent comparative factorial statistics. Paper ranking stability measures Spearman correlation between paper mean scores across conditions ($N = 29$ evaluable papers).

Panel A: Paired Score Sensitivity to Identity Removal and Evidence Restriction

Comparison	Paired Cells (N)	Pearson r	MAE	Scientific Interpretation
FULL vs MASKED Text	112	0.9937	0.0089	Tests dependence on explicit publication identity
FULL vs TITLE/ABSTRACT Only	88	0.7274	0.0758	Tests sensitivity to restriction of available paper evidence

Panel B: Title/Abstract-Only Sensitivity by Hypothesis Family

Hypothesis Family	Paired Cells (N)	Pearson r	MAE	Coverage Note
Hypothesis 1 (Predictive Suppression)	30	0.6003	0.1444	5 papers with evaluable title/abstract evidence
Hypothesis 2 (Feedforward Error)	38	0.6776	0.0614	4 papers with evaluable title/abstract evidence
Hypothesis 3 (Ubiquity)	20	1.0000	0.0000	2 papers; insufficient coverage for general inference

Panel C: Zero-Manuscript-Evidence Control (Prior Only)

Input Condition	Calls (N)	LO Mean [$\pm$ SD]	GO Mean [$\pm$ SD]	Context Contrast ($\Delta_{\text{LO-GO}}$)
Intact Evidence (FULL Reference)	18	0.512 ± 0.081	0.474 ± 0.076	+0.038 (Active context differentiation)
Prior Only (Zero Manuscript Evidence)	18	0.450 ± 0.110	0.450 ± 0.108	0.000 (Parity across contexts)

Supplementary Table S10: Evidence Availability, Publication-Identity, and Prior-Only Sensitivity Analyses. De-identifying full manuscript text produced minimal score adjustments (FULL vs MASKED, Panel A), whereas restricting available evidence to titles and abstracts produced larger deviations (FULL vs TITLE/ABSTRACT Only, Panel A) with the largest attenuation in predictive suppression mechanisms (Panel B). In the zero-manuscript-evidence condition (Panel C), models retained baseline affirmative support from the ontology and task framing, but the Local vs. Global Oddball context differentiation observed under intact manuscript evidence was completely absent. Title/abstract-only Hypothesis 3 coverage was limited to 20 paired cells from two papers and is therefore not interpreted as evidence of general invariance across the literature.

Ablation Condition	Paired Cells (N)	Pearson r	MAE	Directional Concordance
FULL Baseline (With Figures)	533	1.000	0.000	100.0% (Baseline)
NO_FIGURE (Figures Stripped)	533	0.8539	0.2253	99.42% (518/521 non-zero cells)

Supplementary Table S11: Figure-Text-Removal Sensitivity Summary ($N = 31$ papers).

Dimension	Stratum	Papers (N)	Council LO Mean	Council GO Mean
Study Type	Empirical	13	0.419	0.428
Study Type	Computational / Theory	18	0.351	0.365
Modality	In Silico	19	0.363	0.376
Modality	Human In Vivo	7	0.357	0.416
Modality	Animal In Vivo	5	0.511	0.424
Publication Era	1982–2014	12	0.368	0.393
Publication Era	2015–2020	7	0.447	0.488
Publication Era	2021–2025	12	0.347	0.309
Publication Status	Peer-Reviewed Journal	30	0.381	0.392
Publication Status	Preprint (Westerberg & Xiong 2025)	1	0.166	−0.244

Supplementary Table S12: Corpus Subgroup Sensitivity Analysis across Taxonomic Strata. Demonstrates that council-estimated evidence support across Local and Global Oddball contexts varies by experimental modality and publication era, with Westerberg & Xiong (2025) exhibiting a distinct negative Global Oddball profile.

Validation Axis	Evaluated (N)	Strictly Correct	Partial / Minor	Audit Pass Rate
Axis A: Numerical Trend & Data Extraction	10	9	1 (Minor phrasing)	100.0% PASS (10/10)
Axis B: Empirical Effect Direction	10	10	0	100.0% PASS (10/10)
Axis C: HPC-36 Ontology Factor Mapping	10	9	1 (Invertebrate)	100.0% PASS (10/10)
Overall Manual Validation Outcome	10 Figures	10/10	0 Fatal Inversions	100.0% PASS

Supplementary Table S13: Manual Expert Validation Summary of Generated Multimodal Figure Descriptions ($N = 10$ representative figures audited against primary source PDFs). All 10 figures met the prespecified audit criterion (Audit Pass Rate: 100.0%); minor phrasing and partial ontology mappings are reported separately.

Context	Hyp.	N	Human Mean	Council Mean	Human–Council [95% CI]	$t(df), p$
Local Oddball (LO)	H_1	21	$+0.537 \pm 0.041$	$+0.337 \pm 0.027$	$+0.200$ [0.111, 0.289]	$t(20) = 4.69, p = 0.0001$
	H_2	23	$+0.552 \pm 0.045$	$+0.363 \pm 0.026$	$+0.189$ [0.117, 0.261]	$t(22) = 5.42, p < 0.0001$
	H_3	14	$+0.520 \pm 0.034$	$+0.268 \pm 0.045$	$+0.252$ [0.161, 0.344]	$t(13) = 5.95, p < 0.0001$
Global Oddball (GO)	H_1	18	$+0.600 \pm 0.048$	$+0.299 \pm 0.031$	$+0.302$ [0.182, 0.422]	$t(17) = 5.32, p < 0.0001$
	H_2	22	$+0.560 \pm 0.043$	$+0.259 \pm 0.049$	$+0.300$ [0.205, 0.396]	$t(21) = 6.54, p < 0.0001$
	H_3	11	$+0.557 \pm 0.065$	$+0.312 \pm 0.057$	$+0.244$ [0.080, 0.408]	$t(10) = 3.32, p = 0.0077$

Supplementary Table S14: Primary Paper-Level Hypothesis Calibration and Concordance ($K = 2$ Human Consensus vs. Autonomous Council). Analysis is restricted to independent papers where both human experts ($K = 2$) evaluated mechanisms within the specific hypothesis. Council models were averaged within paper first. SEM denotes standard error across independent papers ($SD/\sqrt{N}$). Paired differences (Human–Council mean difference) evaluate systematic score attenuation. Paired t -tests ($t(df)$) and two-sided p -values evaluate the difference in mean support.

Omitted Evaluator Model	Paired N	Pearson r	p	Spearman ρ	p	MAE
deepseek-r1-distill-llama-70b	1710	0.9553	< 0.0001	0.9589	< 0.0001	0.0465
gemma-3-27b	1733	0.9747	< 0.0001	0.9472	< 0.0001	0.0408
gpt-oss-claude-4.5-sonnet	1801	0.9920	< 0.0001	0.9844	< 0.0001	0.0086
mistral-nemo-12b-thinking	1775	0.9732	< 0.0001	0.9698	< 0.0001	0.0239
olmo-3-32b-think	1798	0.9900	< 0.0001	0.9800	< 0.0001	0.0122
phi-4-reasoning-plus	1778	0.9825	< 0.0001	0.9735	< 0.0001	0.0255
qwen3-14b-gemini-3-pro	1800	0.9797	< 0.0001	0.9649	< 0.0001	0.0243
qwen3.5-40b-claude-4.5-opus	1724	0.9497	< 0.0001	0.9178	< 0.0001	0.0440

Evaluator profiles with hypothesis-level scores but unassigned individual factor cells:

gemma-4-31b	1816	—	—	—	—	—
gpt-oss-safeguard-120b	1816	—	—	—	—	—

Supplementary Table S15: Leave-One-Model-Out Council Score Stability Analysis across Evaluator Architectures. Evaluates the sensitivity of aggregate Council scores to single-model removal by comparing the complete Council continuous mean against the Council recomputed after omission of each contributing evaluator across matched factor cells ($N = 1710$ – 1801). Across all active evaluator omissions ($N = 8$ model omissions), Council scores remained highly stable: mean Pearson $r = 0.9747$ (95% CI: [0.9619, 0.9874], range 0.9497–0.9920), mean Spearman $\rho = 0.9621$ (95% CI: [0.9441, 0.9800], range 0.9178–0.9844), and mean MAE = 0.0282 (95% CI: [0.0163, 0.0402], range 0.0086–0.0465). Confidence intervals are computed using Student’s t -distribution ($df = N_{\text{omissions}} - 1 = 7$) with model

omission as the unit. MAE is reported in original $[-1.0, +1.0]$ score units. Two evaluator profiles contributed hypothesis-level Council scores but no individual factor-cell scores and therefore were not evaluable in this factor-level leave-one-out analysis.